

MarineGuard AI — Investigation Report

SYNTHETIC DEMONSTRATION DATA — not real-world observations.

1. Incident information

Field	Value
Incident ID	MG-LIVE-20260907
Title	Arabian Sea Active Operational Incident (Live Feeds)
Location	
Observation time	2026-09-07T16:53:06Z
Provenance	SYNTHETIC DEMONSTRATION DATA

2. Satellite observation

SAR scene associated with this incident. Detection mode: **U-Net ML Inference**. Trained U-Net weights were used for segmentation.

3. Spill detection

Field	Value
Mode	U-Net ML Inference
Confidence	90.9%
Status	Completed

4. Spill geometry

Field	Value
Area	69.28 km²
Length	41.08 km
Width	1.69 km
Perimeter	85.48 km
Centroid	19.10004, 71.62822
Shape	elongated

5. Probable origin

Field	Value
Latitude	19.12628
Longitude	71.7801
Estimated origin time	2026-09-07T12:23:06Z
Window	2026-09-07T10:53:06Z → 2026-09-07T13:53:06Z
Uncertainty radius	5.19 km

Probable origin is an analytical backward integration combining surface current vectors and 3% wind leeway with Coriolis deflection. It defines a probable search window, not an exact spill location.

6. Hindcast result

Backward trajectory points: 25. Model: backward_euler_drift_integration.

7. Forecast

Lead time	Latitude	Longitude
+0 h	19.125	71.85
+6 h	19.1233	71.9432
+12 h	19.12159	72.03639
+24 h	19.11818	72.22278
+48 h	19.11136	72.59554

Forward trajectory forecast incorporates predicted hydrodynamic surface current and 3% wind leeway. Uncertainty corridor expands over time due to oceanic turbulence and atmospheric variation.

8. AIS traffic analysis

Field	Value
Total vessels	7
Spatial candidates	4
Temporal candidates	4
Final candidates	3

9. Candidate vessel ranking

Rank	MMSI	Score	Priority
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10. Evidence

11. Confidence

Detection confidence 90.9%. Drift uncertainty radius 5.19 km. Forecast corridor width 23.6 km.

12. Data gaps

- Satellite detection may include false positives, especially in prototype/demo mode.
- AIS coverage may be gapped, delayed, or spoofed.
- Ocean wind/current fields are sparse demonstration vectors, not a circulation model.

13. Responsible-AI disclaimer

This system prioritises vessels for investigation and does not establish definitive legal responsibility. The highest-ranked vessel is a highest-priority candidate only. Results require human validation and must not be used as the sole basis for legal enforcement.